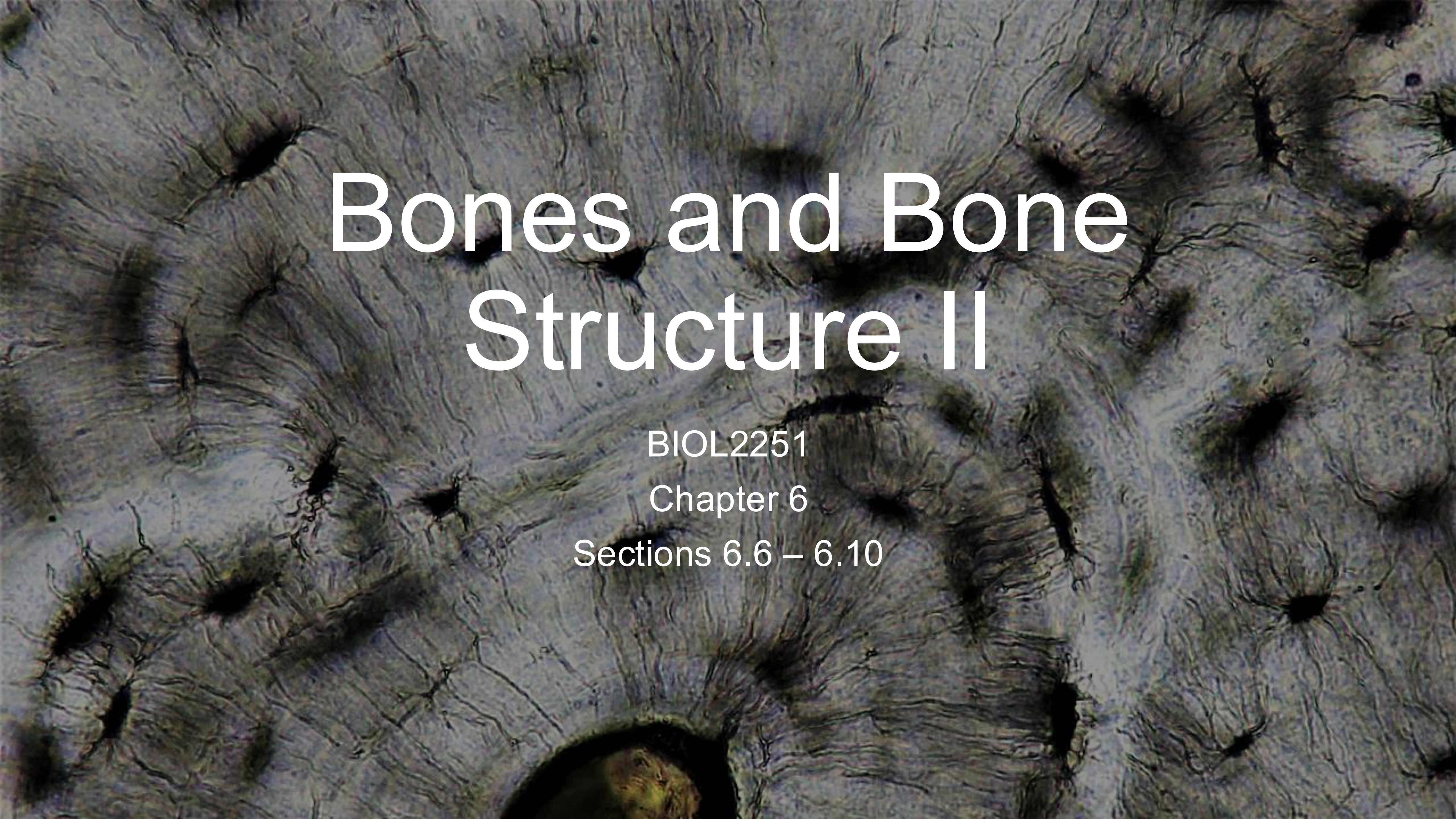


Bones and Bone Structure II

BIOL2251

Chapter 6

Sections 6.6 – 6.10

Bone Remodeling Process

Bone Remodeling Process

- **Instructions**
  - Work in small groups (4-6 people)
  - Create a concept map that shows:
    - What is bone remodeling? (Define it)
    - What three cell types are involved?
    - What does each cell type do during remodeling?
    - What happens when osteoblast and osteoclast activities are balanced?
    - What happens when osteoclasts work faster than osteoblasts?
    - What happens when osteoblasts work faster than osteoclasts?
    - Give examples of situations where the balance tips each way
- **Bloom's taxonomy = Remember, Understand, Analyze**

Bone Remodeling Process

- **Bone remodeling** = continuous recycling of bone matrix
- **Osteocytes** → continuous maintenance, sensing, signaling
- **Osteoclasts** → remove matrix and osteons, release minerals
- **Osteoblasts** → deposit matrix and form osteons, store minerals
- **Balance**
  - **Balanced** → healthy bone maintenance
  - **Osteoclasts > Osteoblasts** → weakening (examples: immobility, aging)
  - **Osteoblasts > Osteoclasts** → strengthening (examples: exercise, growth)

Nutrient and Hormone Matching

Nutrient and Hormone Matching

- **Instructions**
  - Work on your own
  - **Match nutrients/hormones to their role and main effects on bone**
  - Pair to compare and discuss answers
- **Bloom’s taxonomy = Remember, Understand**

Nutrient and Hormone Matching

Items to Match

- Calcium
- Phosphorus
- Vitamin D (Calcitriol)
- Growth Hormone
- Thyroid Hormone
- Sex Hormones
- Parathyroid Hormone (PTH)
- Calcitonin

Effects to Consider

- Part of bone structure
- Helps absorb calcium from food
- Stimulates bone growth
- Raises blood calcium
- Lowers blood calcium
- Closes growth plates

Nutrient and Hormone Matching

Nutrient/Hormone	Main Effect on Bone
Calcium	Building block of bone (makes it hard)
Phosphorus	Building block of bone (works with calcium)
Calcitriol	Helps absorb calcium from food
Growth Hormone	Makes bones grow (longer and thicker)
Thyroid Hormone	Helps bones grow normally
Sex Hormones	Growth spurt + maintain bone density + close growth plates
PTH	Raises blood calcium (breaks down bone if needed)
Calcitonin	Lowers blood calcium (stops bone breakdown)

Exercise and Bone Case Analysis

Exercise and Bone – Case Analysis

- **Instructions**
  - Work in small groups (4-6 people)
  - Each group will receive a specific case study involving different exercise patterns
    - Predict changes to the bone
    - Explain changes in terms of bone’s response to stress
  - Debrief as a class
- **Bloom’s taxonomy = Apply, Analyze, Evaluate**

Exercise and Bone – Case Analysis

- **Case 1 = Astronaut in Space (6 months):** Maria exercises 2 hours daily on a treadmill in space (weightless environment).
  - What happens to her bone density?
  - Why does this happen?
- **Case 2 = Tennis Player:** Jake (right-handed) has played competitive tennis for 10 years.
  - How do his right and left arm bones compare?
  - What causes this difference?

Exercise and Bone – Case Analysis

- **Case 3 = Bedridden Patient:** After surgery, Sarah is in bed for 3 months (cannot walk).
  - What happens to her leg bones?
  - Why is early movement important?

- **Case 4 = Swimmer vs. Runner:** Both athletes train intensely for 10 years:
  - Lisa: Competitive swimmer
  - Maria: Distance runner
  - Who has denser bones? Why?

Exercise and Bone – Case Analysis

Key Principle: "Bone adapts to the stress placed on it"

- **Stress on bone** → bone gets stronger/denser
- **No stress on bone** → bone gets weaker/less dense
- **"Use it or lose it"**

How It Works:

- Osteocytes sense mechanical stress on bone
- **High stress** → signal osteoblasts to build more bone
- **Low/no stress** → signal osteoclasts to remove bone (why maintain expensive tissue that's not needed?)

Exercise and Bone – Case Analysis

CASE 1: ASTRONAUT IN SPACE
What happens: Bone density decreases

Why:
- **No gravity = no mechanical stress**
  - Even though she exercises, there's no weight/impact force
- Osteocytes detect "no stress"
- Osteoclasts break down bone (body doesn't maintain what it doesn't need)
- Osteoblasts decrease activity

Exercise and Bone – Case Analysis

CASE 2: TENNIS PLAYER

- **Comparison:**
- **Right arm:** higher density/thicker bones
- **Left arm:** Normal bone density

Why the difference:

- **Right arm:** Constant mechanical stress from hitting ball
    - Impact forces
    - Muscle pulling on bone
    - Osteocytes sense high stress
    - Osteoblasts build more bone
- **Left arm:** Normal daily use (no extra stress)
    - No special stimulus to build extra bone

Exercise and Bone – Case Analysis

CASE 3: BEDRIDDEN PATIENT
What happens to leg bones: Rapid bone loss

Why:
- **No weight-bearing = no mechanical stress**
  - Legs normally support body weight and movement
- Osteocytes detect absence of stress
- Osteoclasts very active (breaking down bone)
- Osteoblasts decrease activity
- Body thinks: "Don't need strong leg bones if not using them"

Why early movement important:
- **Prevents excessive bone loss**

Exercise and Bone – Case Analysis

CASE 4: SWIMMER VS. RUNNER

- **Who has denser bones:** RUNNER (Maria)
- **Bone Density Comparison:**
- Runner > Swimmer

Why the difference:

Runner:

- **High-impact, weight-bearing exercise**
- Ground reaction forces with each step
- Osteocytes sense high mechanical stress
- Osteoblasts very active → build dense bone

Swimmer:

- **Non-weight-bearing exercise** (water buoyancy supports body)
- No impact forces (water absorbs forces)
- Minimal mechanical stress on bones
- Some benefit from muscle pulling on bone, but not much
- Osteocytes sense low stress

Important Lesson:

- **Type of exercise matters for bone health**

Calcium Homeostasis Flowchart

Calcium Homeostasis Flowchart

- **Instructions**
  - Form 12 total groups (with mini-whiteboards)
  - Draw flowcharts showing how the body responds to different levels of calcium
    - Must include: what detects change, hormones released, organs affected, and final result
  - Groups will share and we will debrief as a class
- **Bloom's taxonomy = Remember, Understand, Apply, Analyze**

Calcium Homeostasis Flowchart

Scenario A

- Blood calcium drops to 7.5 mg/dL (normal: 8.5-10.5)
  - What detects this?
  - What hormone(s) released?
  - What organs respond?
  - What's the result?

Scenario B

- Blood calcium rises to 12 mg/dL (normal: 8.5-10.5)
  - What detects this?
  - What hormone(s) released?
  - What organs respond?
  - What's the result?

Calcium Homeostasis Flowchart

SCENARIO A: LOW BLOOD CALCIUM

FLOWCHART:
1. DETECTION:
- **Parathyroid glands** detect low calcium
- Located on back of thyroid in neck

2. HORMONE RESPONSE:
- **PTH (Parathyroid Hormone) RELEASED**

3. TARGET ORGANS & RESPONSES:
BONES:
- PTH stimulates osteoclasts (indirectly)
- Osteoclasts break down bone
- **Result:** Calcium released from bone into blood (FAST - minutes to hours)

KIDNEYS:
- PTH tells kidneys to retain calcium (less lost in urine)
- PTH tells kidneys to activate vitamin D
- **Result:** Save calcium; make active vitamin D

INTESTINES (indirect via vitamin D):
- Active vitamin D increases calcium absorption from food
- **Result:** More calcium absorbed into blood (SLOWER - hours to days)

4. FINAL RESULT:
- Blood calcium returns to normal
- PTH secretion decreases (negative feedback)

Calcium Homeostasis Flowchart

SCENARIO B: HIGH BLOOD CALCIUM FLOWCHART:

1. DETECTION:

- **Thyroid gland (C cells)** detect high calcium

2. HORMONE RESPONSE:

- **Calcitonin RELEASED**
- **PTH secretion STOPS**

3. TARGET ORGANS & RESPONSES:

BONES:

- Calcitonin inhibits osteoclasts
- Bone breakdown stops
- **Result:** Less calcium released from bone

KIDNEYS:

- Calcitonin increases calcium excretion in urine
- **Result:** More calcium eliminated from body

INTESTINES:

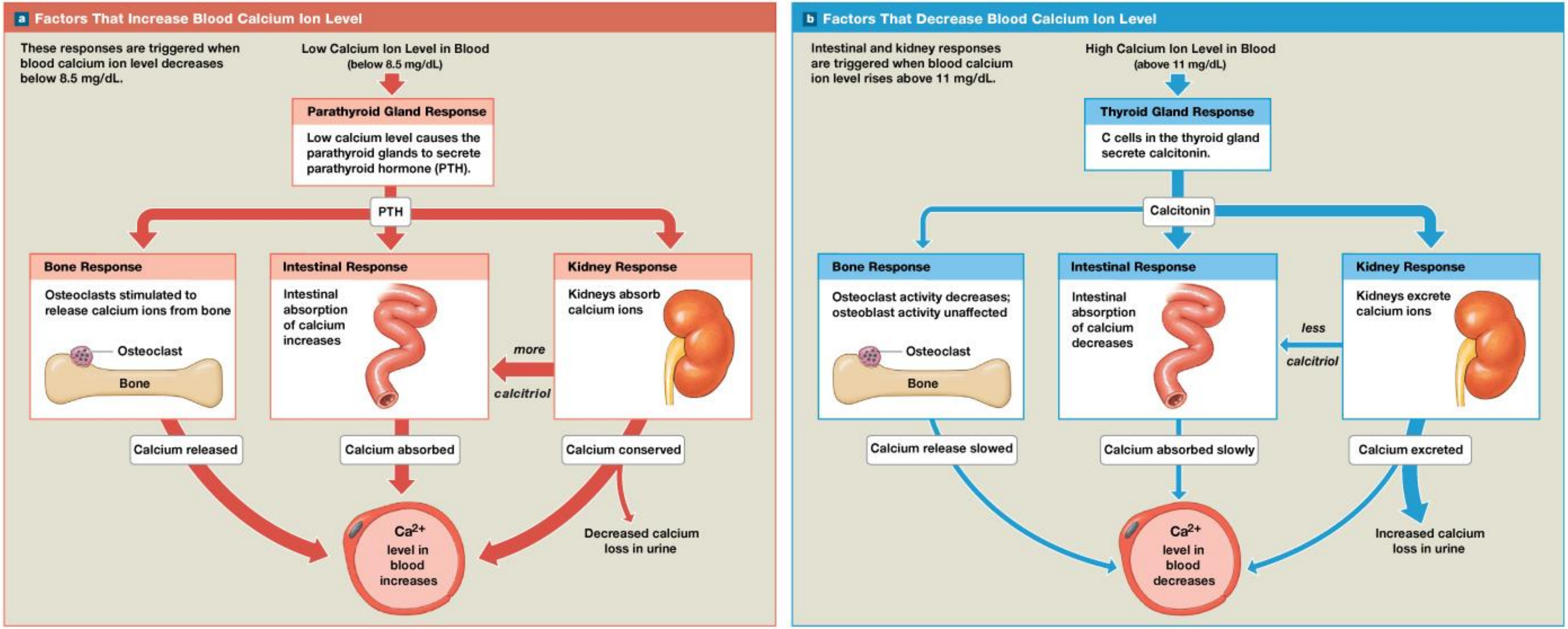
- Less calcitriol activation (no PTH signal)
- **Result:** Less calcium absorbed from food

4. FINAL RESULT:

- Blood calcium returns to normal
- Calcitonin secretion decreases
- PTH returns to baseline

Calcium Homeostasis Flowchart

Condition	Detector	Hormone Up	Main Actions	Result
LOW Ca^{2+}	Parathyroid glands	PTH ↑ Calcitriol ↑	Break down bone; kidneys save Ca^{2+}; absorb more Ca^{2+} from food	Raises blood Ca^{2+}
HIGH Ca^{2+}	Thyroid (C cells)	Calcitonin ↑	Stop bone breakdown; kidneys eliminate Ca^{2+}	Lowers blood Ca^{2+}



a Factors That Increase Blood Calcium Ion Level
These responses are triggered when blood calcium ion level decreases below 8.5 mg/dL.
Low Calcium Ion Level in Blood
(below 8.5 mg/dL)
Parathyroid Gland Response
Low calcium level causes the parathyroid glands to secrete parathyroid hormone (PTH).
PTH
Bone Response
Osteoclasts stimulated to release calcium ions from bone
Osteoclast
Bone
Intestinal Response
Intestinal absorption of calcium increases
Kidney Response
Kidneys absorb calcium ions
more
calcitriol
Calcium released
Calcium absorbed
Calcium conserved
Decreased calcium loss in urine
Ca2+ level in blood increases
b Factors That Decrease Blood Calcium Ion Level
Intestinal and kidney responses are triggered when blood calcium ion level rises above 11 mg/dL.
High Calcium Ion Level in Blood
(above 11 mg/dL)
Thyroid Gland Response
C cells in the thyroid gland secrete calcitonin.
Calcitonin
Bone Response
Osteoclast activity decreases; osteoblast activity unaffected
Osteoclast
Bone
Intestinal Response
Intestinal absorption of calcium decreases
Kidney Response
Kidneys excrete calcium ions
less
calcitriol
Calcium release slowed
Calcium absorbed slowly
Calcium excreted
Increased calcium loss in urine
Ca2+ level in blood decreases

Fracture Repair Sequencing

Fracture Repair Sequencing

- **Instructions**
  - Work on your own
  - Sequence the four stages of bone fracture repair
    - **Explain why each stage is necessary**
  - Pair with a partner to discuss answers
  - Debrief as a class
- **Bloom’s taxonomy = Understand, Apply**

Fracture Repair Sequencing

Stages to Sequence

- Hematoma formation
- Spongy formation
- Bone remodeling
- Hard callus formation

Questions for Each Stage:

- What happens?
- Why is this stage necessary?

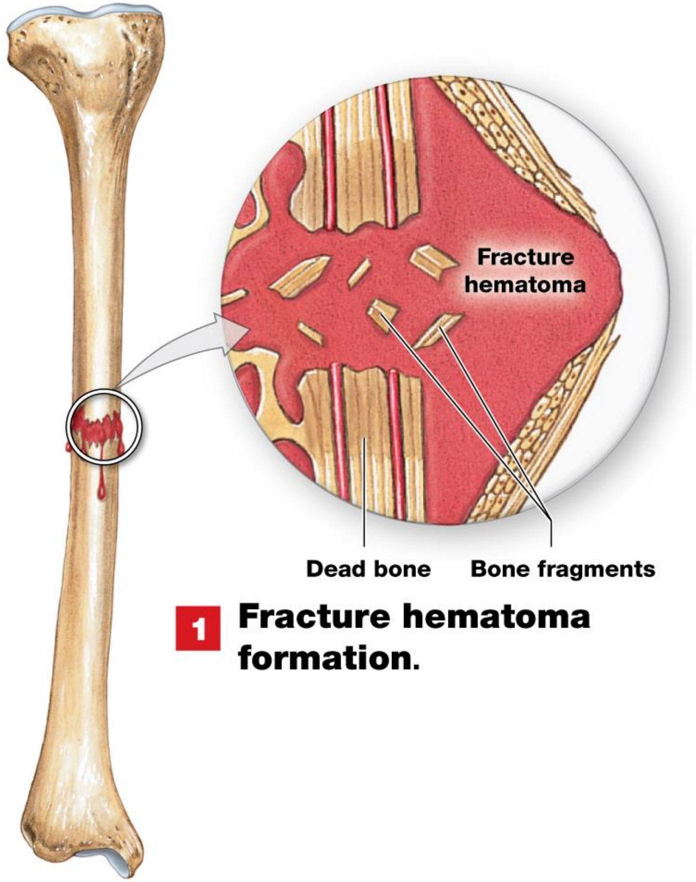
Stage 1: Hematoma Formation

What happens:

- Bone breaks → blood vessels tear → bleeding
- Blood clot (hematoma) forms at fracture site
- Fills gap between broken bone ends
- Bone tissue near fracture dies (no blood supply)
- Swelling, pain, inflammation begin

Why necessary:

- Blood clot provides scaffold for healing
- Stabilizes fracture initially



1 Fracture hematoma formation.

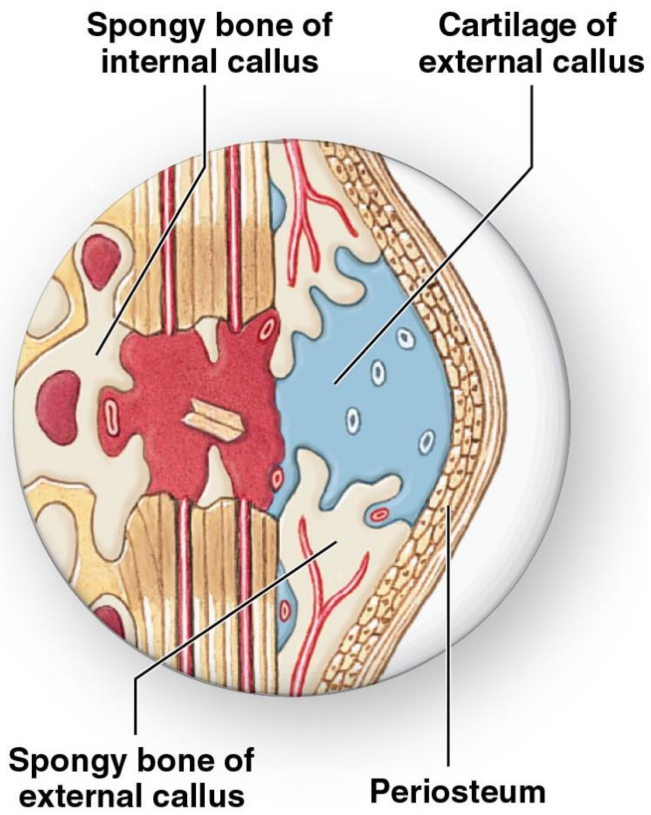
Stage 2: Callus Formation

What happens:

- Fibroblasts make collagen fibers
- Soft callus forms (fibrocartilage + connective tissue)
- Callus bridges gap between bone ends (internal and external)
- Soft callus (cartilage) is replaced by bone
- Internal callus = spongy bone that unites the fracture
- External callus = cartilage and bone stabilizes outer edges of the fracture

Why necessary:

- Stabilizes fracture better than blood clot
- Bridges gap between bone ends



2 Callus formation.

Stage 3: Spongy Bone Formation

What happens:
- Osteoblasts replace central cartilage with spongy bone
- Callus hardens (mineralizes)
- Bony callus bridges fracture completely
- Fracture becomes stable enough to support weight

Why necessary:
- Replaces weak cartilage with spongy bone
- Restores structural integrity

3 Spongy bone formation.

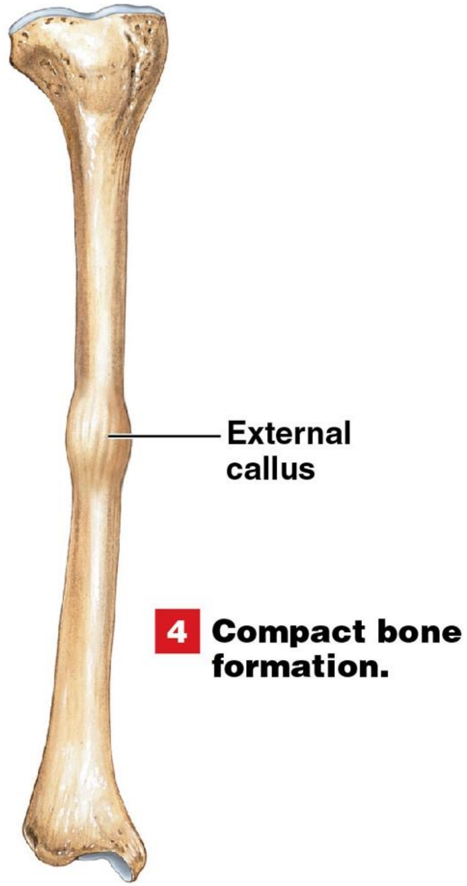
Stage 4: Bone Remodeling

What happens:

- Spongy bone replaced with compact bone (where needed)
- Bone reshaped to match original structure
- Trabeculae align along stress lines
- Medullary cavity restored
- Fracture line may disappear on X-rays
- Final bone often stronger than before!

Why necessary:

- Restores normal bone shape
- Optimizes bone strength
- Removes excess weight
- Returns bone to pre-fracture (or better) strength



Bone Health Case Studies

Bone Health Case Studies

- **Instructions**
  - Work in small groups (4-6 people)
  - You will receive a case study where you must:
    - **Analyze bone health**
    - **Identify problems**
    - **Explain what is happening**
    - **Propose solutions to the problem**
  - Groups will volunteer to share, followed by a class debrief
- **Bloom’s taxonomy = Analyze Evaluate**

Bone Health Case Studies: Case 1

- **Postmenopausal Woman**
- Linda, age 60, went through menopause at age 52. She doesn't exercise, drinks wine each night, and avoids dairy (but doesn't take supplements). Her mother had a hip fracture at age 70.
- Questions:
  - What's happening to her bones at the cellular level?
  - What are her major risk factors?
  - What should she do?

Bone Health Case Studies: Case 2

- **Young Female Athlete**
- Sarah, age 20, is a competitive long-distance runner (60+ miles/week). She hasn't had a period in 18 months. She avoids fats and dairy. She just got a stress fracture.
- Questions:
    - Why did an athlete get a stress fracture?
    - What hormonal problem exists?
    - What's happening to her bones?
    - What must she change?

Bone Health Case Studies: Case 3

Elderly Man Post-Hip Fracture

- Robert, age 75, broke his hip 6 weeks ago. He's been in bed during recovery. He has diabetes and has a poor appetite. His blood calcium is low (7.8 mg/dL). His fracture is healing slowly.
- Questions:
  - Why is his blood calcium low?
  - Why is fracture healing slow?
  - What does he need for better healing?